

EcoGuardian is a state-of-the-art environmental monitoring platform that bridges the gap between **concerned citizens**, **administrative oversight**, and **field authorities**. Leveraging **real-time synchronization via Socket.io** and **Gemini 1.5 Flash AI**, every environmental threat — from illegal logging to wildlife injuries — is tracked, triaged, and resolved with total transparency.

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USER ROLES

6

INCIDENT TYPES

4

SEVERITY LEVELS

8

REPORT STATUSES

AI

GEMINI POWERED

PROBLEM & SOLUTION

- ✗

Environmental incidents go unnoticed due to fragmented systems
- ✗

Citizens lack a transparent channel to verify complaint resolution
- ✗

Authorities struggle to triage real vs. fake/spam reports
- ✗

No unified real-time coordination between all stakeholders

- ✓

Unified platform with role-based dashboards for all parties
- ✓

End-to-end transparent timeline from submission to resolution
- ✓

Gemini AI automatically flags suspicious and fake reports
- ✓

Live Socket.io push notifications keep everyone in sync

KEY FEATURES



Real-time Ops Center

Live incident tracking with instant push notifications via Socket.io



AI Sentinel Insights

Gemini 1.5 Flash analyses images and verifies report authenticity



Evidence Protocol

Authorities upload photo-of-resolution as visual proof to reporters



Eco-Cloud Intercom

Real-time peer-to-peer chat for on-ground field coordination



Gamified Leaderboard

Points & community rankings reward active environmental reporters



Offline Resilience

Graceful fallback to in-memory storage if database is unreachable

TECHNOLOGY STACK

FRONTEND

React 19

TypeScript

Vite 6

Framer Motion

Tailwind CSS 4

Lucide Icons

BACKEND

Node.js

Express.js

Socket.io 4

MongoDB

Mongoose

TypeScript

AI & SERVICES

Gemini 1.5 Flash

Google GenAI

Twilio SMS

SYSTEM ARCHITECTURE



AI Pipeline: Image → Gemini classifies **type & severity** & recommends action · Report → AI returns **isReal, confidence %** & analysis · AI generates inspiring **community summaries** for the dashboard

TRI-ROLE ARCHITECTURE



Citizen (User)

Submit & track environmental reports
Upload photo evidence with AI scan
Earn points · View leaderboard



Command (Admin)

AI authenticity verification of all reports
Assign incidents to field authority
Manage alerts & broadcast announcements



Field Authority

Accept & investigate assigned cases
Upload photo-of-resolution as proof
Real-time chat coordination

INCIDENT TYPES & SEVERITY



Land Damage



Illegal Logging



Injured Wildlife



Animal Road Accident



Pollution



Other

Low

Medium

High

Critical

REPORT LIFECYCLE



Pending



Verified



Assigned



Investigating



Solved



Resolved

or



Rejected

WHY ECOGUARDIAN STANDS OUT



AI image classification + fake-report detection using Gemini 1.5 Flash



Photo-of-Resolution — visual proof for every resolved case



Full real-time stack — every status change pushed instantly to all dashboards



Gamification rewards citizens with points for each verified report



Demo Mode — runs completely offline, ideal for presentations

FUTURE ROADMAP

- 🌱

Predictive Anomaly Heatmaps with ML trend forecasting
- 🦋

Direct NGO Integration for immediate wildlife rescue dispatch
- 📄

Exportable Environmental Governance PDF Reports
- 📱

Mobile application (React Native) for field authorities